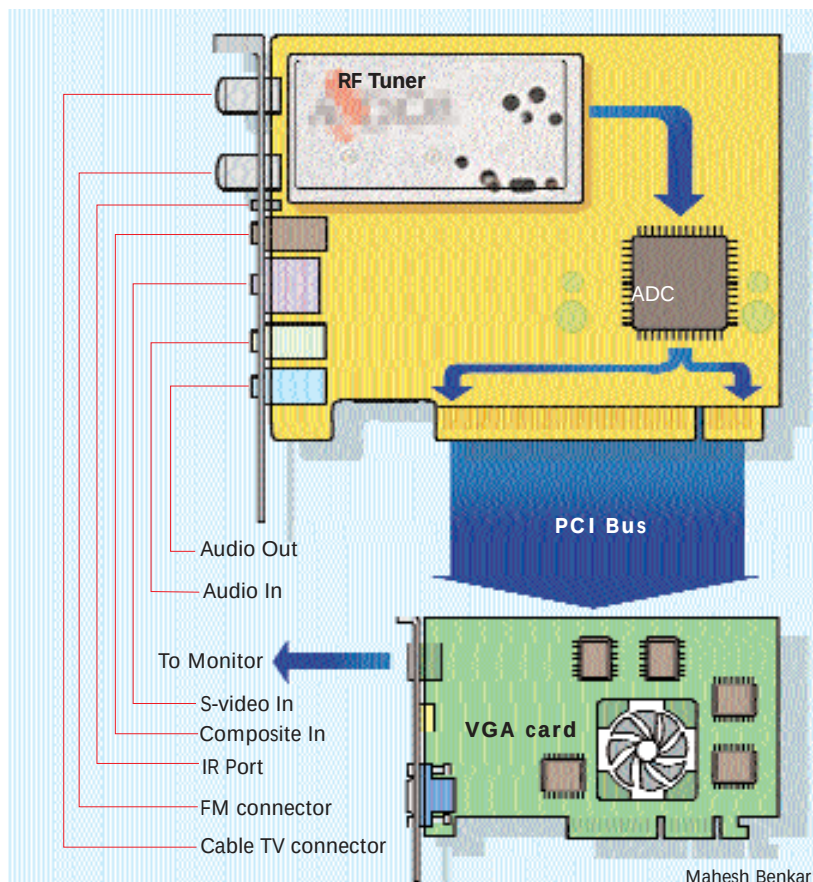


What is a TV Tuner?

A TV-Tuner card is just another PCI card. Most cards have a very simple layout with a RF (Radio Frequency) Tuner unit and an ADC (Analog To Digital Converter) chip. The ADC chips are mostly manufactured by Brooktree (Conexant) or Philips; and most tuner units are made by Philips. The RF tuner unit is just like the one you have in your TV. The frequency of TV signals lie between 40 MHz and 300 MHz for VHF, and about 800 MHz for UHF. FM radio bandwidth is also between 88 MHz and 109 MHz, hence the name RF.

The cable wire is plugged in to the Tuner unit, where the RF signal is separated from the carrier signal and then passed to the ADC. The converter then samples this data, depending on the signal—either NTSC or PAL. The data is then passed on to the video card, where it is overlaid on the frame-buffer. This saves a lot on the system resources. However, capturing content will definitely hog system resources, and affect any other tasks you may be doing. This includes playing games, or working with multimedia applications. Working in Word or Excel, or surfing the Net while capturing content shouldn't be a problem.



Mahesh Benkar

Heads Up

Using a TV tuner isn't really as simple as plugging it in, loading the drivers and sitting back, and forgetting about it. There are certain precautions to be taken. While doing this test, we came across three things that you should keep an eye out for:

- Those with VIA chipset motherboards, with the onboard Savage 3D video controller, need to remember that the Savage 3D chips have an overlay problem, because the older drivers aren't DirectDraw compliant. Get a driver fix from the manufacturers Web site, or your hardware is likely to ruin your captured footage.
- RealTek ALC-650 onboard chips can distort captured sound, but help is at hand. Download the latest drivers and a patch from the RealTek Web site to fix this problem.
- Lastly, and most importantly, remember that your TV cable carries a small electrical charge along with your favourite programmes. This charge is small enough to ignore, but if it surges, you might have yourself with a fried PC. Unfortunately, there's nothing much you can do, as only your local cable guy has control over his wires. All you can do is cross your fingers, pray a lot and disconnect the cable during storms.

good remote control. Most PixelView cards had the drivers, its native application called PlayTV and Cyberlink PowerVCR on the same CD. The Pixelview PlayTV XP was the only exception—it has WinDVR and Powerdirector SE.

Compro was the best product as far as software bundles were concerned—they gave both Ulead VideoStudio and Ulead DVD Movie Factory; the Compro DVD software looks suspiciously like Cyberlink PowerDVD—it has the same configuration features. A sound editing tool would have been a welcome addition. Pixelview also supplied all the necessary wires, and the drivers and software came in one CD. Mercury needs to gather its wits on the packaging front—a detailed manual is necessary to explain the driver installation and software configuration, since they don't bundle any third-party utilities. Even the remote doesn't offer much in terms of usability, but is really sleek.

Ease of use

A product that's really easy to use is obviously more desirable. Driver and application crashes during use, reduced points here. The Pinnacle software configuration is one of the best. It can be configured easily; all we had to do was follow the instructions that the wizard gave. When a problem occurred, the wizard told us exactly what was wrong. Once the configuration is